

Nan Wang

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Phone: +1 (236)965-2522

Email: nan.wang@sauder.ubc.ca

Education

University of British Columbia 2021 – 2027 (expected)

PhD in Strategy and Business Economics

Committee: Guillermo Marshall (co-chair), Michael Sullivan (co-chair), Ekaterina Khmelnitskaya, Chunhua Wu

University of Wisconsin-Madison 2019 – 2021

MSc in Economics

Beijing Institute of Technology 2015 – 2019

BComm in Accounting

BA in Economics (dual)

Research Interests

Industrial Organization, Platform Economy, Machine Learning, Digital Economy

Working Papers

Platform Competition and Multi-homing: Evidence from Ride-hailing (Job Market Paper)

Abstract Markets organized by two-sided platforms tend to be concentrated. Yet the effect of adding platforms may differ from the standard one-sided intuition, because of platforms' two-sidedness and widespread multi-homing. I study how competition affects platforms' two-sided price structure and market efficiency, and how multi-homing changes these effects. Using novel data on the universe of ride-hailing orders, I develop an empirical equilibrium model of rider demand, driver labor supply, and platform matching technology.

Algorithmic Price Discrimination (with Yang Yu)

Abstract Modern pricing algorithms exploit high-dimensional consumer information, making price discrimination subtle and difficult for consumers and regulators to detect. We propose a machine-learning-based method to test whether a given piece of consumer information is used to price-discriminate. Built on a double/debiased machine-learning framework, the method removes bias from model misspecification and overfitting. We apply it to test whether a ride-hailing platform charges frequent riders differently from occasional riders. A naive linear analysis produces persistent false positives, whereas our method finds no statistically significant difference in prices across riders.

Dual Role on Shoulders: Who Benefits from Platform Exclusivity?

Abstract A vertically integrated digital platform plays a dual role: it operates a marketplace for third-party products while selling its own first-party products on that marketplace. This paper examines whether such a platform should keep its first-party products exclusive to its own marketplace, and

what this controversial practice implies for welfare. Using a model of Nash bargaining and Bertrand competition, I show that giving up exclusivity raises platform profits but lowers consumer surplus. The rationale is that selling first-party products on rival platforms serves as a collusive device: it softens competition for platform usage and ultimately leads to higher product prices.

Teaching Experience

Instructor

COMM 295 Managerial Economics

2026 Summer

Teaching Assistant

COMM 491 Strategic Management in Business

2023 Fall

Research and Professional Experience

DiDi

Pricing Researcher

Beijing, China

Oct 2024 – Dec 2025

Research Assistant

Prof. James Brander

2022 – 2023

Prof. Ekaterina Khmelnitskaya

2022 – Present

UBC IO Workshop

Organizer

2022 – 2024

Conferences and Summer Schools

2025: Luohan Academy Annual Digital Economy Conference

2024: Econometric Society Summer School in Dynamic Structural Econometrics (DSE)

2023: Competition & Regulation European Summer School and Conference (CRESSE)

Awards and Fellowships

President's Academic Excellence Initiative PhD Award

2021 – 2026

Dean Earle D MacPhee Memorial Fellowship in Commerce and Business Administration

2021 – 2026

Sauder School of Business Graduate Award

2021 – 2022

C K Choi Fellowship in Business Administration

2021

Other Information

Programming: Python, Julia, SQL, R

Language: English (Fluent), Mandarin (Native)

References

Guillermo Marshall

Professor of Strategy and Business Economics
Sauder School of Business
University of British Columbia
guillermo.marshall@sauder.ubc.ca

Ekaterina Khmelnitskaya

Assistant Professor of Strategy and Business Economics
Sauder School of Business
University of British Columbia
ekaterina.khmelnitskaya@sauder.ubc.ca

Chunhua Wu

Associate Professor of Marketing
Sauder School of Business
University of British Columbia
chunhua.wu@sauder.ubc.ca

Michael Sullivan

Assistant Professor of Strategy and Business Economics
Sauder School of Business
University of British Columbia
michael.sullivan@sauder.ubc.ca

James Brander

Professor of Strategy and Business Economics
Sauder School of Business
University of British Columbia
james.brander@sauder.ubc.ca